Reward Design Transferability Under Climate Shift: Evaluating MARL Energy Controllers in Dubai Desert Weather

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***Abstract*—Multi-agent reinforcement learning (MARL) has demonstrated strong results for urban energy management in the CityLearn benchmark, yet all existing work trains and evaluates policies under the same climate distribution. This paper investigates zero-retraining climate transfer from a temperate US residential district to Dubai desert conditions using the CityLearn 2023 six-building dataset and real TMYx weather profiles. We compare Independent PPO (IPPO) and Multi-Agent PPO (MAPPO) across three reward designs: flat shared, local individual and Dubai-weighted shaped reward, and introduce three analytical contributions. The Reward Transfer Score (RTS) quantifies the degree to which the reward-to-KPI mapping is preserved under climate shift. Seasonal regime analysis identifies when and why policies fail under different Dubai climate regimes. KPI conflict Pareto frontier analysis characterises how the cost–discomfort trade-off itself is restructured under deployment. Results across three training seeds reveal a consistent pattern. Cost and carbon emissions decrease under Dubai evaluation while thermal discomfort increases, indicating that policies adapt to reduce energy consumption but fail to adapt to intensified cooling demand. The Reward Transfer Score is approximately zero (-0.0226, $p = 0.9662$), confirming that reward change is statistically decoupled from operational KPI change under transfer. Pareto analysis shows that MAPPO preserves the cost–discomfort geometry approximately $3.16\times$ better than IPPO, while IPPO with local individual reward achieves the most balanced overall KPI profile and lowest training variance. These findings demonstrate that the central challenge of climate transfer is not mere performance degradation but a structural shift in the optimization trade-off: a reward function calibrated in one climate continues to be optimised in the deployment climate but now corresponds to an undesirable cost–discomfort operating point. Reliable deployment therefore requires reward functions calibrated to the target climate alongside environment-aware training strategies.**

***Index Terms*—Multi-agent reinforcement learning, CityLearn, climate transfer, reward shaping, KPI optimisation, building energy management, demand response, Dubai**

I. Introduction

Urban energy management has become a high impact control problem because the building sector accounts for approximately one-third of global energy use, while HVAC systems alone can consume around half of residential building energy [1], [2]. In multi-building districts, this demand is no longer just a building-level concern. The challenge is therefore not simply to minimize energy use, but to manage an inter-related multi-objective trade-off across cost, comfort, emissions, and load stability. This is precisely the kind of setting in which urban energy management requires scalable decision-making rather than fixed rule-based control.

MARL has emerged as a natural framework for this setting because each building can be modeled as an autonomous agent that observes local conditions, selects actions, and contributes to district outcomes [3]. The CityLearn platform has played a major role in standardizing this line of research by providing a common multi-building simulation environment and a shared KPI-based evaluation protocol [4], [5]. Within this benchmark, policy-gradient methods have become especially important because they can handle continuous control directly, which is essential for building energy systems with batteries, domestic hot water storage, and cooling devices [6]. Recent benchmark results have established Independent PPO (IPPO) and Multi-Agent PPO (MAPPO) as strong representatives of decentralized training and centralized training paradigms, respectively [7], [8].

However, a key limitation of existing CityLearn-based studies is that policies are trained and evaluated under the same climate distribution, leaving cross-climate generalization largely unexplored. As a result, current evidence says little about what happens when a controller trained in one climate is deployed directly in another without retraining. This is not a minor detail. Prior work outside the exact CityLearn transfer setting has already shown that reinforcement-learning control performance and reward sensitivity can vary substantially across climatic contexts [9]. Related transfer-learning work such as MERLIN has shown that policies can sometimes generalize across buildings, but climate transfer in cooperative MARL for urban energy control remains largely unexplored [10].

This paper studies that gap directly. We investigate zero-retraining climate transfer from the CityLearn 2023 US training environment to a Dubai desert weather setting. The central question is not only whether performance changes, but whether the *meaning* of the reward changes.

For this purpose, we compare IPPO and MAPPO under three reward designs: flat shared reward, local individual reward, and Dubai-weighted shared reward. The study is guided by four research questions:

- **RQ1.** Which KPIs change most under zero-shot climate transfer, regardless of reward design?
- **RQ2.** Does reward shaping reduce the transfer-induced degradation or trade-off distortion compared with a flat shared baseline?
- **RQ3.** Is the effect of reward design different for decentralized IPPO and centralized-training MAPPO?

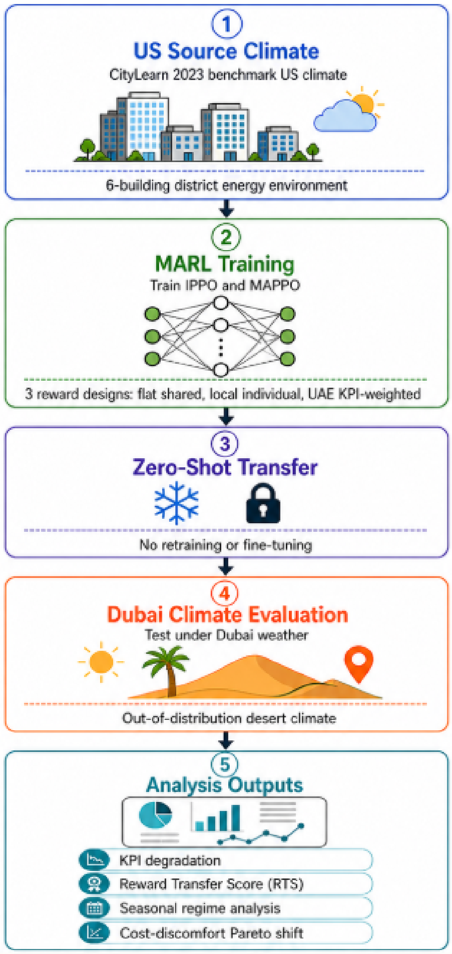

Fig. 1. Overview Of The Proposed MARL Reward Design Transferability Study

- **RQ4.** Does climate transfer alter the structure of the cost-discomfort trade-off frontier?

The main contributions of the paper are as follows:

- A controlled zero-retraining MARL climate-transfer study on the CityLearn 2023 six-building dataset, evaluating IPPO and MAPPO across three reward designs in a Dubai desert weather setting.
- The **Reward Transfer Score (RTS)**, a metric that measures whether reward change remains aligned with KPI change under climate transfer.
- A **seasonal regime analysis** that identifies when transfer fails, when it saturates, and when it still generalizes during the year.
- A **cost–discomfort Pareto analysis** that shows how the operational trade-off geometry shifts under transfer rather than only how average KPI values change.

To provide a clear overview of the experimental setup and evaluation process, Fig. 1 illustrates the overall workflow of the proposed framework.

II. Background and Related Work

Recent work in urban energy control highlights that MARL performance is strongly influenced not only by the choice of algorithm, but also by reward design, evaluation protocols and environmental conditions [4]. The CityLearn platform has enabled standardized benchmarking across multiple buildings and KPIs, showing that different algorithms exhibit varying trade-offs between cost, emissions, comfort and load stability [5]. Policy-gradient methods such as Independent Proximal Policy Optimisation (IPPO) and Multi-Agent PPO (MAPPO) have been widely adopted due to their capacity to handle continuous control and multi-agent coordination in realistic building energy scenarios [7]. However, existing studies mainly focus on performance within a fixed environment setup, with limited analysis of how reward formulations behave under changing conditions.

From a reward design perspective, prior work shows that most approaches rely on scalarized reward functions with fixed weights, which often fail to capture complex trade-offs between competing objectives. Improper reward configuration can lead to suboptimal coordination or unstable behavior across agents. In addition, recent survey findings emphasize that MARL systems are highly sensitive to environment-specific factors and policies trained under one setting may not generalize well to others.

Despite these advances, a key gap remains in understanding how reward design interacts with environmental changes in multi-agent settings. Existing work on transfer and generalization has largely focused on single-agent control or different simulation environments, without systematically studying reward behavior in MARL under modified conditions. This motivates the focus of this project, which examines how different reward formulations influence system behavior under increased cooling demand and how these trade-offs change when the environment becomes more challenging.

A. CityLearn Environment and the 2023 Dataset

CityLearn is an open-source OpenAI Gym-compatible simulation platform specifically designed to evaluate reinforcement learning strategies for demand response and urban energy management [4]. Introduced by Vázquez-Canteli et al. [4], the environment models a district of buildings connected through a shared electricity network, where each building possesses its own energy demand profile and a set of controllable systems. A defining feature of CityLearn is that it provides a standardised and reproducible evaluation framework using five canonical KPIs: electricity cost, carbon emissions, ramping, load factor and thermal comfort. This consistency has made it the dominant benchmark platform for MARL research in the building energy domain, with successive CityLearn Challenges formalising competitive evaluation under identical conditions [5].

CityLearn operates in two main control modes. In its default decentralised mode, the environment returns one reward and one observation per building at each timestep, making it directly compatible with MARL frameworks. In centralised mode, it returns a single aggregate reward and a joint state vector, which can be used to train single-agent baselines or centralised critics. The platform supports customisable reward functions, user-defined datasets, and modular building components including heat pumps, batteries, domestic hot water systems and electric vehicle chargers. These properties give researchers precise control over experimental conditions while preserving physical realism derived from pre-simulated EnergyPlus or Modelica building loads [4].

This work uses the **citylearn_challenge_2023_phase_3_1** dataset, which represents a district composed of six buildings with heterogeneous characteristics. Each building includes components such as space cooling systems, domestic hot water demand, non-shiftable electrical loads and battery storage. The dataset also provides time-series data including outdoor temperature, solar irradiance, electricity pricing and carbon intensity, typically derived from standard weather formats such as EnergyPlus Weather (EPW) files [4].

The environment operates in discrete timesteps, where agents observe the state of each building and apply continuous control actions. This setup allows the evaluation of both individual and coordinated control strategies in a realistic and consistent simulation setting.

B. MARL Paradigms: DTDE and CTDE

Multi-Agent Reinforcement Learning (MARL) provides a principled framework for modelling distributed control in building energy systems, where each building is treated as an independent agent that observes a local state, selects actions and contributes to overall district-level performance [3]. Two principal training paradigms have emerged in the literature, each with distinct computational and coordination properties.

The first paradigm is Decentralized Training and Decentralized Execution (DTDE), also referred to as Independent Learning [3]. In DTDE, each agent learns its policy independently using only its own local observations, treating all other agents as part of a non-stationary environment. A typical example is Independent Proximal Policy Optimization (IPPO), which applies the single-agent PPO algorithm [6] to each building agent in isolation. IPPO maintains a separate actor and critic network per agent, computes advantage estimates using Generalised Advantage Estimation (GAE) from local rollouts and updates parameters using the clipped surrogate objective to prevent excessively large policy updates. Because training is fully decentralised, IPPO is operationally attractive in privacy-sensitive settings where buildings cannot share internal data during learning. Empirically, IPPO has demonstrated strong and consistent performance across multiple CityLearn KPIs, often achieving competitive results with lower variance than centralised alternatives [7]. Its primary limitation is the non-stationarity problem: from any individual agent's perspective, the environment changes as all other agents simultaneously update their policies, which can impede convergence when inter-agent coupling is strong. This approach is simple and often leads to stable learning, but it does not explicitly model coordination between agents.

The second paradigm is Centralized Training with Decentralized Execution (CTDE). In CTDE, agents are trained using shared global information to produce richer value estimates and encourage coordinated behaviour, while still acting independently using only local observations at deployment time. Multi-Agent Proximal Policy Optimisation (MAPPO), proposed by Yu et al. [7], is the leading CTDE method in urban energy MARL. MAPPO extends IPPO by replacing the per-agent local critic with a single centralised critic that conditions on the concatenated observations of all agents. This richer input enables the critic to capture inter-agent dependencies and provide a less noisy training signal for each actor. Yu et al. [7] demonstrated that, when combined with appropriate training practices including value normalisation, critic input design, and gradient clipping, MAPPO achieves highly competitive performance in cooperative multi-agent games. In the CityLearn context, the top-performing solution in the NeurIPS 2022 CityLearn Challenge employed MAPPO as its core algorithm, demonstrating the advantage of centralised critic coordination in district-level energy management [5]. However, MAPPO is also more sensitive to training instability: the centralised critic introduces higher variance in some settings, and its performance can degrade more noticeably under challenging or out-of-distribution conditions [11].

Previous work has shown that IPPO often achieves strong average performance with stable learning behavior, while MAPPO can provide better coordination in some cases but is more sensitive to training conditions [7].Comparative evaluations in the CityLearn setting have consistently shown that both paradigms offer complementary strengths. Costea [11] conducted a structured taxonomy-driven evaluation comparing IPPO, parameter-shared IPPO and MAPPO across residential CityLearn buildings, with separate KPI reporting for summer and winter periods. The seasonal breakdown revealed that MAPPO's ramping penalty increased substantially during winter relative to other methods, a finding that would be entirely invisible in an annualised average and that underscores the importance of multi-scenario evaluation. Gorsane et al. [12] further highlighted that algorithm rankings in cooperative MARL are highly sensitive to evaluation protocols, particularly the choice of performance metric and the number of independent training seeds, reinforcing the need for statistically rigorous comparisons across both DTDE and CTDE approaches.

C. Reward Design in Cooperative MARL

Reward design is one of the most critical yet underexplored dimensions of MARL for urban energy systems, as it directly defines the objective that agents optimise during training and governs how competing operational goals are balanced [13]. In multi-objective building energy control, the reward must simultaneously account for electricity cost, carbon emissions, occupant thermal comfort and grid stability, objectives that frequently conflict with one another. The choice of reward structure has substantial and often independent effects on individual KPIs, coordination behaviour and overall system performance.

Three main reward paradigms appear in the literature. The first is a flat shared reward, where all agents receive an identical signal derived from aggregated district-level performance. This formulation encourages cooperative behaviour by aligning all agents with a common objective, but it obscures individual building contributions and can slow credit assignment, making it harder for each agent to determine whether its specific actions improved or degraded performance. The second is a local individual reward, where each building agent is evaluated solely on its own observable KPIs. This simplifies credit assignment and can accelerate single-building optimisation, but risks producing selfish policies that fail to account for district-level coordination, particularly with respect to grid ramping and peak demand reduction [14]. The third paradigm, and the most prevalent in recent work, is the shaped scalarised reward, where multiple objectives are combined as a weighted sum. The CityLearn baseline reward, for example, combines discomfort, electricity, ramping, and solar self-consumption penalties with equal weights of 0.1 each. While this allows explicit control over system priorities, it introduces the challenge of weight selection: small changes in objective weights can lead to significantly different policy outcomes and KPI trade-offs [14].

An important practical consequence is the reward-KPI translation problem, in which the cumulative reward achieved during training does not reliably predict performance across all evaluation metrics [4]. Dhamankar et al. [13] established this empirically in one of the first systematic comparisons of MARL algorithms on CityLearn, showing that policies with similar cumulative returns can diverge substantially in their individual KPI profiles. [15] further demonstrated that the specific choice of training objective significantly affects both energy consumption outcomes and coordination behaviour in residential MARL settings. These findings collectively suggest that reward design should be treated as a primary research variable rather than a secondary implementation detail.

In this project, three reward formulations are evaluated: a flat shared reward with equal weights (serving as the paper baseline), a local individual reward where each building optimises only its own signal, and a Dubai-weighted shared reward in which comfort and cost are assigned higher importance (0.35 and 0.30 respectively) to reflect the operational priorities of a high-cooling-demand environment. This design allows a controlled comparison of how reward structure influences system behaviour under both standard and intensified thermal conditions.

D. Transfer and Generalisation in Building Energy RL

Transfer and generalisation are central challenges for deploying reinforcement learning controllers in real buildings, where policies must remain effective under conditions that differ from the training environment. In practice, building energy systems are subject to shifting climates, heterogeneous building stocks, changing occupancy patterns, and evolving tariff structures. All of these can cause a policy trained under fixed simulation assumptions to degrade or fail when these assumptions are violated. This creates a clear reality gap between benchmark performance and deployment readiness, even when simulation results appear strong.

Generalisation in urban energy MARL is multi-dimensional. Cross-climate generalisation is perhaps the most operationally significant dimension, as heating and cooling loads, solar availability, and thermal inertia vary substantially across geographic regions. Yu et al. [9] explicitly examined this issue by combining RL-based HVAC control with urban climate modelling, demonstrating that policy transferability and reward sensitivity are strongly affected by climatic context. Their findings indicate that climate-diverse evaluation should be incorporated as a standard component of benchmarking rather than treated as an optional extension. Cross-building generalisation presents a related challenge: buildings differ in envelope characteristics, storage capacity, HVAC configuration and occupant behaviour, making building-specific training from scratch both expensive and impractical for community-scale deployment.

Some recent work has explored transfer learning in this domain. The MERLIN framework, proposed by Nweye et al. [10], directly addresses cross-building transfer in CityLearn by combining multi-agent offline learning with policy transfer across a grid-interactive residential community. Using smart meter data, MERLIN demonstrated that policies trained on one building can be transferred to others with comparable performance to policies trained from scratch on building-specific data, thereby reducing training cost. However, the study also showed that building-specific load and occupancy profiles continued to affect final policy quality, indicating that transfer can reduce but not eliminate the need for adaptation. In the CityLearn Challenge 2022, Nweye et al. [5] further showed that a single shared-parameter policy network could generalise across buildings of different characteristics when trained with data augmentation and ensemble imitation learning, providing a practical pathway toward community-wide deployment.

A third dimension is offline-to-online generalisation. Because unconstrained online exploration in real buildings can disrupt comfort or damage equipment, recent work has explored learning from historical operational data before limited online adaptation. This approach is directly relevant to deployment scenarios where data collection is constrained and safety requirements are strict.

However, most existing studies focus either on single-agent systems or on fixed environment conditions. There is limited work that jointly considers multi-agent learning, reward design and environmental changes within the same framework. In particular, the interaction between reward design and generalisation remains underexplored. Policies trained under one reward formulation may behave differently when the environment changes, especially in multi-agent settings where coordination is required.

This gap motivates the focus of this project, which examines how reward design influences system behavior under modified climate conditions and how different learning approaches respond to these changes.

Overall, the literature shows that MARL has become a dominant approach for urban energy control due to its ability to model distributed decision-making across multiple buildings. Platforms such as CityLearn have played a key role in enabling standardized evaluation, allowing different algorithms to be compared under common KPIs such as cost, emissions, comfort and load stability. Within this setting, policy-gradient methods like IPPO and MAPPO have demonstrated strong performance, with decentralized approaches offering stability and centralized methods enabling coordination.

At the same time, reward design emerges as a critical factor in shaping system behavior, as it directly defines how competing objectives are balanced. Existing studies highlight that small changes in reward formulation can lead to significantly different outcomes, especially in cooperative multi-agent settings. In addition, recent work emphasizes that MARL performance is highly sensitive to environmental conditions and that policies trained in one scenario may not generalize well to others.

Despite these advances, existing work does not explicitly analyze how reward formulations behave under distributional shifts such as climate transfer. In particular, the relationship between reward design and KPI outcomes under out-of-distribution conditions remains insufficiently understood. This gap motivates the present study, which isolates the effect of climate change on reward–KPI alignment in a multi-agent setting.

III. Methodology

A. CityLearn Environment Setup

This work uses the CityLearn dataset, specifically the *citylearn_challenge_2023_phase_3_1* scenario [5]. The dataset represents a small urban district composed of six buildings, each with different energy demand profiles and consumption patterns.

The dataset includes time-series data such as outdoor temperature, humidity, solar irradiance, electricity pricing and carbon intensity. In addition, each building has its own internal variables, including cooling demand, domestic hot water demand, non-shiftable load and storage state of charge.

These variables are updated at each timestep and allow the environment to simulate realistic building behavior over time. This makes it possible to observe how control decisions affect both energy consumption and thermal comfort.

The CityLearn environment simulates the interaction between control actions and building energy systems over time. At each timestep, the environment provides observations that describe the current state of each building. These include indoor temperature, energy demand values, storage levels and external weather conditions.

The observation space combines both internal building variables and external environmental signals. This means that each agent must make decisions by balancing building needs, such as maintaining comfort, with external factors such as weather conditions and electricity cost.

Control actions are continuous and applied per building. These include battery charging and discharging, domestic hot water control and cooling system operation. Each action influences the future state of the building, meaning that decisions have long-term effects rather than immediate outcomes.

The environment follows a step-based interaction process. At each timestep, agents receive observations, take actions, and then receive the next state along with a reward signal. This process continues over a full simulation horizon, allowing the agents to learn policies that optimize performance over time.

In the initial stage, the environment was implemented using a centralized setup, where all building observations were combined into a single vector and controlled using a single PPO agent. While this simplified the implementation, it did not reflect the distributed nature of real-world systems.

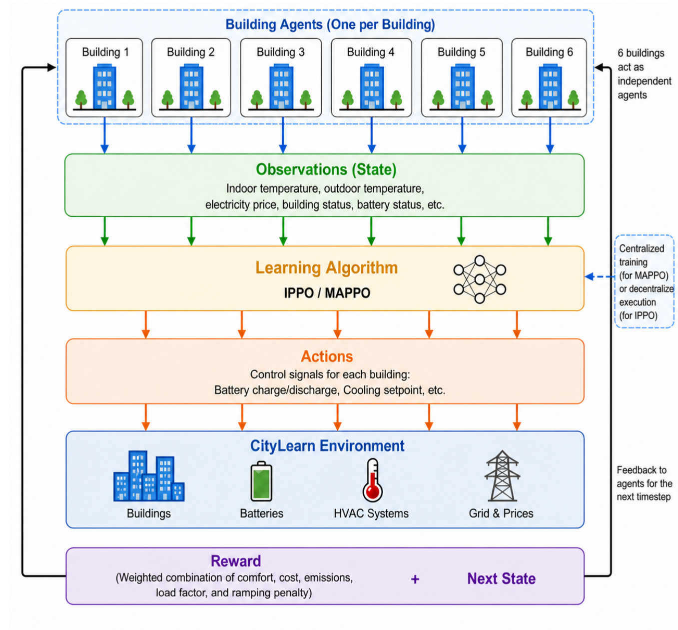

Fig. 2. CityLearn multi-agent environment setup. Each building acts as an independent agent that observes its local state, takes control actions, and interacts with the shared environment.

In the updated setup, the environment is modeled as a multi-agent system, where each building is treated as an independent agent with its own observation and action space. Agents make decisions based only on local observations, which better reflects real-world operation where buildings act independently.

For methods such as MAPPO, a centralized critic is used during training. In this case, a global state is constructed by combining observations from all buildings. This allows the model to capture interactions between buildings while still maintaining decentralized execution during deployment. The overall interaction process, including observations, actions and reward computation, is illustrated in Fig. 2.

To study different system behaviors, multiple reward formulations are considered. A shared reward is used to encourage cooperation across buildings by optimizing district-level objectives. A local reward assigns each building its own objective, allowing agents to act independently. In addition, a weighted reward is introduced to place more emphasis on comfort and cost, reflecting the requirements of a cooling-dominated environment.

To evaluate zero-shot climate transfer, the original CityLearn weather file was replaced with real TMYx 2007–2021 weather data for Dubai International Airport, obtained from the EnergyPlus weather database. This source provides climate data designed for building simulation in EPW format, which makes it appropriate for building energy evaluation studies [?].

The Dubai evaluation used the June–August peak summer period to represent extreme cooling conditions. In this period, the Dubai weather data had a mean outdoor temperature of 36.0°C and a maximum of 47.0°C, compared with the US training climate mean of 28.0°C and maximum of 38.1°C.

This was not a synthetic fixed-temperature shift. Instead, the weather file was replaced using real hourly climate data. Outdoor temperature, relative humidity, solar irradiance, and their predicted values were updated, while all other building parameters, including thermal load profiles, HVAC sizing, battery capacity, and pricing, were kept constant. This setup isolates the effect of climate shift and allows the evaluation to focus on how reward design affects system behavior under more extreme cooling demand.

B. Algorithm Selection and Justification

This study compares two MARL algorithms: Independent Proximal Policy Optimization (IPPO) and Multi-Agent Proximal Policy Optimization (MAPPO). These algorithms are selected because they provide a clean comparison between decentralized and centralized training paradigms while remaining within the same PPO-based algorithm family. This avoids mixing fundamentally different learning mechanisms and allows the study to focus on whether centralized training improves or weakens robustness under climate transfer.

IPPO is used as the decentralized learning baseline [16]. In IPPO, each building agent maintains an independent policy and learns from its own local observations and reward signal. This corresponds to a DTDE setting, where agents do not rely on a centralized critic during training.

MAPPO is used as the centralized-training counterpart [7]. MAPPO follows the CTDE paradigm. During training, MAPPO uses a centralized critic that can access aggregated information from all building agents, allowing the critic to estimate the value of the joint system state. During execution, however, each building still selects actions using only its local policy. This makes MAPPO suitable for testing whether access to global information during training improves coordination and transfer robustness, or whether it increases sensitivity when the policy is evaluated under unseen Dubai weather conditions.

Both algorithms are compatible with the CityLearn multi-agent setting because each building can be modeled as one agent, each agent receives building-level observations, and each agent outputs continuous control actions for energy-related devices such as cooling, electrical storage, and domestic hot water storage.

Together, IPPO and MAPPO provide a controlled comparison between decentralized and centralized MARL while keeping the underlying optimization family consistent. IPPO tests whether independent building-level learning is more robust under climate shift, while MAPPO tests whether centralized training improves coordination or increases sensitivity to out-of-distribution weather [7], [8].

C. IPPO Architecture and Training Procedure

Independent Proximal Policy Optimization (IPPO) is used as the decentralized baseline. Each building is treated as an independent agent with its own actor–critic model. Agent i receives only its local observation vector o_i and outputs a continuous action vector a_i. The policy and value function are written as

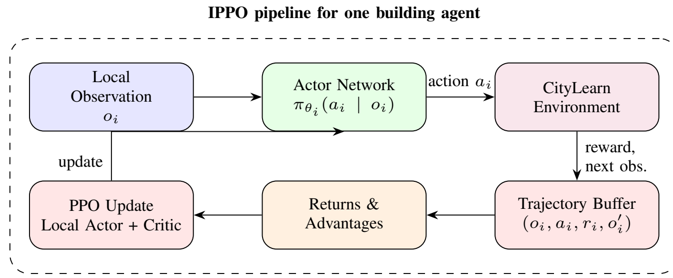

Fig. 3. IPPO training pipeline in CityLearn for one building agent.

$$\pi_{\theta_i}(a_i \mid o_i), \qquad V_{\phi_i}(o_i), \tag{1}$$

where θ_i and ϕ_i denote the actor and critic parameters.

In the implemented architecture, each actor uses a squashed Gaussian policy. The actor predicts the mean of a Gaussian distribution, learns a log-standard deviation vector, samples a raw action, and applies a $\tanh$ squashing function so that outputs remain in the normalized interval $[-1, 1]$. The critic is a separate multilayer perceptron that estimates the scalar value of the local observation. Both actor and critic use two hidden layers of 256 units with Tanh activations.

During training, each agent stores its normalized observations, sampled actions, log-probabilities, rewards, and value estimates over a rollout of 1,024 steps. Advantages are then computed independently for each agent and used to update the policy with the PPO clipped objective

$$\begin{aligned} L_i^{\text{IPPO}}(\theta_i) = E_t\Big[&\min\big(r_{i,t}\hat{A}_{i,t}, \\ &\text{clip}(r_{i,t}, 1-\epsilon, 1+\epsilon)\,\hat{A}_{i,t}\big)\Big]. \end{aligned} \tag{2}$$

Training uses 10 update epochs per rollout, minibatches of size 256, a learning rate of 3×10^{-4}, gradient clipping, entropy regularization, and observation normalization. The main characteristic of IPPO is that learning remains fully decentralized, making it a useful baseline for testing whether independent building-level policies transfer more robustly to Dubai climate

D. MAPPO Architecture and Training Procedure

Multi-Agent Proximal Policy Optimization (MAPPO) is used as the centralized training counterpart. MAPPO follows the centralized-training/decentralized- execution paradigm. Each building still acts using its own local policy, but a centralized critic is used during training. For each agent i, the actor is

$$\pi_{\theta_i}(a_i \mid o_i), \tag{3}$$

while the critic estimates the value of the global district state

$$V_\phi(s_t), \tag{4}$$

where

$$s_t = [o_{1,t}, o_{2,t}, \ldots, o_{N,t}] \tag{5}$$

is the concatenation of all building observations.

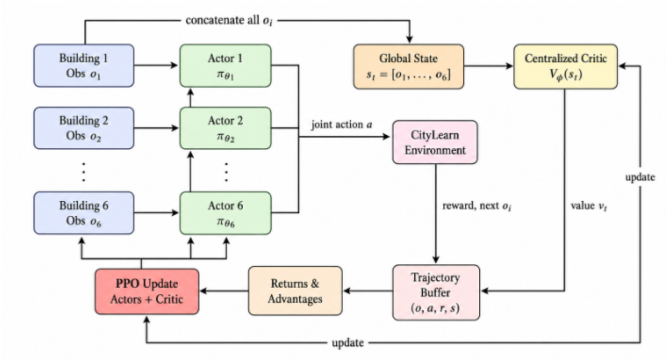

Fig. 4. MAPPO training pipeline for 6 building for CityLeran Environment

The actor architecture is the same as in IPPO: a squashed Gaussian policy with two hidden layers of 256 units and Tanh activations. The centralized critic is also a multilayer perceptron with the same hidden size, but it receives the joint district state instead of a single building observation.

During rollout collection, each building first computes its normalized local observation and samples an action using its own actor. The centralized critic evaluates the concatenated state of all buildings. The environment executes the joint action, returns rewards, and the resulting trajectories are used to update both the decentralized actors and the centralized critic. In the implemented version, the critic is trained using the mean team reward across buildings, so MAPPO learns from a cooperative district-level training signal.

The actor update again follows the PPO clipped objective,

$$L^{\text{MAPPO}}(\theta) = E_t\Big[\min\big(r_t\hat{A}_t,\\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\,\hat{A}_t\big)\Big]. \tag{6}$$

where the advantages are derived from the centralized critic. As in IPPO, training uses 1,024-step rollouts, 10 update epochs, minibatch size 256, entropy regularization, gradient clipping, and observation normalization.

The main distinction between MAPPO and IPPO is therefore the critic structure. MAPPO can exploit cross-building information during training, which may improve coordination, but it may also become more sensitive to distribution shift under zero-shot climate transfer.

E. Reward Design

Reward design is a central part of this study because it determines what the agents are encouraged to prioritize during training. Instead of optimizing a single objective, the reward combines four practical signals that capture important aspects of building energy control: thermal discomfort, electricity import, ramping and solar-related self-consumption behavior. In simple terms, the reward penalizes agents when comfort is not maintained, when more electricity is imported from the grid, when demand changes too abruptly between timesteps, and when grid electricity is used even though solar generation is available.

All three reward designs in this paper use these same four components. The difference lies in how the reward is assigned and how strongly each component is weighted. This keeps the comparison focused on reward structure rather than on changes to the learning algorithm itself.

The first reward is the **flat shared** design. Here, the four components are weighted equally and combined into a single district-level reward that is shared by all buildings. This encourages cooperation because every agent is rewarded according to the overall district outcome. However, it also makes credit assignment harder, since an individual building cannot clearly separate its own contribution from the behavior of the rest of the district.

The second reward is the **local individual** design. It uses the same equal weighting, but each building receives its own local reward based only on its own behavior. This gives each agent a clearer training signal and makes the problem more decentralized, although it may reduce coordination across buildings.

The third reward is the **Dubai-weighted** design. This is again a shared district-level reward, but the weights are shifted to place slightly more importance on thermal comfort and electricity import, while reducing the relative emphasis on ramping. The purpose of this design is to reflect the fact that in a hot, cooling-dominated climate such as the Dubai, maintaining comfort is expected to become more critical than in the original training climate.

An important design decision is that electricity cost and carbon emissions are not directly optimized as reward terms. Instead, they are treated as evaluation KPIs. This makes the reward-to-KPI alignment question meaningful, because a policy that performs well according to its training reward may still behave poorly when judged by deployment-oriented system metrics.

F. Climate Transfer Protocol

The climate-transfer experiment follows a strict zero-retraining setting. All policies are trained only on the original CityLearn 2023 US climate and are then evaluated in Dubai climate without any further learning or adaptation. This means that the trained policy parameters remain fixed and the same controller is used directly in the new climate.

To keep the comparison controlled, only the weather data are changed. The building definitions, observation and action spaces, pricing signals, carbon intensity, and all other dataset elements remain unchanged. In practice, the Dubai evaluation environment is created by copying the original CityLearn dataset and replacing only the `weather.csv` file. This ensures that any performance change can be attributed mainly to the climate shift rather than to other modifications in the environment.

The Dubai weather profile is based on the public TMYx 2007–2021 record for Dubai International Airport. This provides a realistic hot-weather evaluation setting rather than an artificial temperature perturbation. As a result, the experiment better reflects a real deployment question: how well would a controller trained in one climate perform if it were applied directly in a much hotter one?

TABLE I
TRAINING CONFIGURATION (IDENTICAL FOR IPPO AND MAPPO).

Parameter	Value
Total environment steps	250,000
Rollout length	1,024
PPO update epochs	10
Minibatch size	256
Discount γ	0.99
GAE λ	0.95
Clip range ϵ	0.20
Entropy coefficient	0.01
Value-loss coefficient	0.50
Max gradient norm	0.50
Learning rate (Adam)	3×10^{-4}
Hidden size (MLP, 2 layers)	256
Activation	tanh
Action squashing	tanh on Gaussian samples
Action safety factor η	0.75
Observation normalisation	Welford running mean/var, clip ± 10
Random seeds	{42, 2, 7}

G. Training Configuration

Training was carried out in two stages. A short 100k-step pilot was first used during development to verify that the multi-agent environment wrapper, reward functions, logging pipeline and training loops were operating correctly. These pilot runs were used only for implementation validation and are not included in the final reported analysis.

All results reported in this paper come from the final 250k-step runs. Each algorithm–reward combination was trained from scratch using three independent random seeds. The same hyperparameter configuration was used for both IPPO and MAPPO so that the comparison remained fair. Table I summarizes the full setup.

Observation normalization was applied during training so that variables with very different numerical scales, such as temperature, demand, and storage state, could be handled more consistently by the neural networks. In addition, an action safety factor was used when mapping policy outputs into the native CityLearn action space. This keeps actions within a conservative operating range and avoids unrealistic control behavior during training.

For the final paper, all main results are reported as mean $\pm$ standard deviation across the three completed seeds, rather than from a single run, in order to reduce sensitivity to stochastic variation in MARL training.

H. Experimental Matrix

The full experiment is organized as a $2 \times 3 \times 2$ design. The first dimension is the learning algorithm, with IPPO and MAPPO. The second dimension is the reward design, with flat shared, local individual, and Dubai-weighted rewards. The third dimension is the evaluation climate, where each trained policy is tested both in the original US climate and in the Dubai transfer climate.

This produces six training conditions per seed: two algorithms multiplied by three reward designs. Once a model is trained, it is evaluated in both climates without changing its weights. Repeating the same process across three seeds makes it possible to analyze both average performance and consistency.

This matrix is designed to answer the main question of the paper: whether climate transfer changes the relative behavior of different reward designs, and whether this effect depends on the MARL training paradigm.

I. Evaluation Protocol

After training, each policy is evaluated in a deterministic setting for one full simulation year (8,760 steps) under both climates, but since the environment supports only up to 2,208 steps, the program terminates after that. During evaluation, no exploration noise is added. This ensures that the comparison reflects the behavior of the learned controller itself rather than random action sampling.

Two kinds of outputs are recorded. The first is the accumulated reward, which shows how well the policy performs according to its training objective. The second is a set of district-level KPIs provided by CityLearn: total cost, carbon emissions, discomfort proportion, ramping average and daily one-minus-load-factor average. These KPIs are the more important evaluation measure because they describe the actual operational behavior of the system.

To make interpretation easier, performance is also compared against the neutral CityLearn baseline, which represents the system without a learned controller. This helps show whether a trained policy improves on passive behavior or simply optimizes its own reward without delivering better system outcomes.

For the climate-transfer analysis, the main quantity of interest is the percentage change in each KPI when moving from the US climate to the Dubai climate:

$$\Delta_k = \frac{\mathrm{KPI}_k^{\mathrm{Dubai}} - \mathrm{KPI}_k^{\mathrm{US}}}{\mathrm{KPI}_k^{\mathrm{US}}} \times 100\%. \tag{7}$$

A positive value means that the KPI became worse under transfer, while a negative value means that it improved. These per-KPI changes are then combined into summary measures such as average degradation and reward change, which are used later to study reward-to-KPI alignment under climate shift.

All experiments were conducted under identical hyperparameter settings across algorithms to ensure a fair comparison. Reporting mean and standard deviation across multiple seeds reduces sensitivity to stochastic training effects and improves result reliability.

IV. ANALYTICAL FRAMEWORK

This study is not limited to reporting reward and KPI values before and after transfer. Its goal is to explain *why* climate transfer changes policy behavior and whether the reward signal remains meaningful in deployment. To do this, we introduce three analytical tools. First, the Reward Transfer Score (RTS) measures whether reward change remains aligned with KPI change under zero-shot climate transfer. Second, seasonal regime analysis reveals when transfer fails and when it still generalizes. Third, Pareto frontier analysis shows how the cost-discomfort trade-off itself is altered under Dubai deployment. Together, these analyses support the central claim of the paper: a policy trained under a reward specification calibrated in the source climate does not merely lose performance in the Dubai setting, but instead optimizes a cost-discomfort trade-off that no longer matches the deployment reality.

A. Reward Transfer Score

Existing evaluation practice in MARL energy research often reports accumulated reward and operational KPIs separately, which makes it difficult to determine whether the reward signal remains informative once a trained policy is deployed in a new climate. To make this question explicit, we define the *Reward Transfer Score* (RTS), which measures the degree to which the reward-to-KPI relationship is preserved under zero-shot climate transfer.

For each experimental condition i, defined by one algorithm reward-design pair, we compute the relative reward change

$$\Delta r_i = \frac{R_i^{\text{Dubai}} - R_i^{\text{US}}}{\left|R_i^{\text{US}}\right|} \times 100\%, \tag{8}$$

where R_i^{US} and R_i^{Dubai} are the mean agent rewards under the US training climate and the Dubai transfer climate, respectively.

We also compute a composite KPI change by averaging the percentage change across the five district-level CityLearn KPIs:

$$\Delta\bar{k}_i = \frac{1}{|K|}\sum_{k\in K} \frac{\text{KPI}_{k,i}^{\text{Dubai}} - \text{KPI}_{k,i}^{\text{US}}}{\left|\text{KPI}_{k,i}^{\text{US}}\right|} \times 100\%, \tag{9}$$

where K is the set of the five district-level KPIs used in the evaluation, namely `cost_total`, `carbon_emissions_total`, `discomfort_proportion`, `ramping_average`, and `daily_one_minus_load_factor_average`.

The RTS is then defined as the Pearson correlation between the reward-change vector and the composite-KPI-change vector across all experimental conditions:

$$\text{RTS} = \text{corr}\big(\{\Delta r_i\}, \{\Delta\bar{k}_i\}\big)\,. \tag{10}$$

An RTS close to $+1$ indicates strong alignment: reward and KPI change move together under transfer. An RTS near 0 indicates that the reward is largely uninformative about how operational KPIs behave in the target climate. A negative RTS indicates misalignment, meaning that reward change and KPI change tend to move in opposite directions.

In this study, the overall RTS is -0.0226 with $p = 0.9662$. Although the coefficient is close to zero, this result is significant because it indicates a complete decoupling between reward optimization and actual system performance under climate transfer. It shows that, under Dubai transfer, reward change is effectively decoupled from KPI change. In other words, the reward signal cannot be trusted as a proxy for deployment performance in the new climate. This is precisely the kind of reward–KPI misalignment that the paper set out to test.

TABLE II
PER-CONDITION REWARD CHANGE AND COMPOSITE KPI CHANGE UNDER US→DUBAI TRANSFER.

Algorithm	Reward Mode	Δ Reward (%)	Δ KPI Composite (%)
IPPO	Flat Shared	-405.7	-7.94
IPPO	Local Individual	$+0.54$	-6.67
IPPO	Dubai-Weighted	-392.5	-7.09
MAPPO	Flat Shared	-483.1	-3.99
MAPPO	Local Individual	-12.3	-3.87
MAPPO	Dubai-Weighted	-444.6	-3.45

Table II reports the per-condition reward and composite KPI changes. A striking feature is that reward changes are very large in magnitude, especially for the shared-reward conditions, while the composite KPI changes are much smaller. This confirms that the reward signal is reacting to the transfer in a way that does not reflect the actual operational trade-off captured by the KPIs.

This result has a direct interpretation for deployment. A controller may still appear to be optimizing its internal reward, yet that reward no longer tells us whether comfort, cost, emissions, and stability are moving in the intended direction. The RTS therefore formalizes a central insight of this paper: under climate transfer, reward degradation and operational degradation are not the same thing.

B. Seasonal Regime Analysis

Reporting averages annually hides when the transfer succeeds and when it fails. Dubai is not uniformly extreme throughout the year, it includes a mild winter, a transition period in spring, a prolonged high heat summer, and a warmer autumn that partially overlaps with the original US training distribution. To capture this temporal structure, we evaluate the transferred policies over four seasonal regimes: Q1 winter, Q2 spring, Q3 summer, and Q4 autumn.

Table III summarizes the regimes, their temperature ranges, the neutral baseline discomfort level, and the main qualitative feature observed in each quarter. The purpose of this decomposition is not only to show average performance by season, but to identify whether climate transfer produces different *types* of failure and success across the year.

The Q3 summer regime is the clearest example of a saturation effect. Discomfort rises to approximately 0.989–0.994 for all trained policies and for the neutral baseline. This means that during peak heat, the learned controllers are no longer able to separate themselves meaningfully from passive control on thermal comfort. The dominant factor is the severity of the thermal load rather than the choice of reward design or MARL algorithm.

Q1 winter reveals a different pattern. This is the regime with the largest policy spread, and several learned policies perform worse than the neutral baseline on both comfort and cost. In particular, the IPPO shared-reward variants increase discomfort relative to neutral, which suggests that control behavior learned under the source distribution can become actively counterproductive under milder Dubai winter conditions.

Q2 spring provides the strongest positive result for MAPPO. In this transition regime, MAPPO with the flat shared reward

TABLE III
SEASONAL REGIME DEFINITIONS AND KEY CHARACTERISTICS FOR THE DUBAI TRANSFER SETTING.

Season	Months	Dubai Temp. Range	Neutral Discomfort	Key Regime Feature
Q1 Winter	Jan–Mar	12–32°C	0.651	Largest policy differentiation across the six trained conditions
Q2 Spring	Apr–Jun	22–42°C	0.990	Transition regime in which MAPPO shows its clearest advantage
Q3 Summer	Jul–Sep	30–47°C	0.994	Discomfort saturation; all policies converge toward the neutral baseline
Q4 Autumn	Oct–Dec	20–38°C	0.896	All trained policies outperform the neutral baseline on comfort

achieves a discomfort proportion of 0.965 compared with 0.990 for the neutral baseline, an improvement of approximately 2.5 percentage points. This is the largest seasonal comfort gain observed for any learned policy and provides concrete evidence that CTDE can improve coordination during moderate but non-saturated cooling demand.

Q4 autumn is the clearest generalization window. All six trained policies outperform the neutral baseline on discomfort, with values between 0.879 and 0.884 compared with 0.896 for neutral control. This is an important positive finding: climate transfer is not uniformly harmful. Instead, there are conditions under which the learned control behavior still provides a real benefit, especially when the target thermal regime overlaps more closely with the source training regime.

Taken together, the seasonal analysis identifies four qualitatively distinct regimes: winter degradation, spring coordination advantage, summer saturation, and autumn partial generalization. This temporal structure is one of the most informative findings of the study, because it shows that climate-transfer failure is not a single uniform effect but a season-dependent process.

C. KPI Conflict and Pareto Frontier Analysis

The KPI-level results reveal a consistent directional pattern under Dubai transfer which is that cost and carbon emissions tend to decrease, while discomfort increases. This means that transfer does not simply make every KPI worse. Instead, it shifts the system toward a different operating point in which lower cost is obtained alongside poorer comfort. To analyze this more directly, we study the cost-discomfort trade-off using Pareto frontier analysis.

For each experimental condition, we place the US-climate and Dubai-climate outcomes in the two-dimensional space defined by `cost_total` and `discomfort_proportion`. We then compute the Euclidean displacement between the two operating points:

$$d_i = \sqrt{\left(c_i^{\text{Dubai}} - c_i^{\text{US}}\right)^2 + \left(u_i^{\text{Dubai}} - u_i^{\text{US}}\right)^2}, \qquad (11)$$

where c_i denotes cost and u_i denotes discomfort for condition i.

Table IV reports the US and Dubai positions, the discomfort shift, and the Euclidean displacement for all six conditions. The first clear finding is that every condition moves toward lower cost but higher discomfort. This means the frontier does not simply deteriorate in a uniform way. Instead, the trade-off itself is restructured under transfer.

TABLE IV
PARETO FRONTIER DISPLACEMENT IN COST–DISCOMFORT SPACE UNDER US→DUBAI TRANSFER.

Algorithm	Reward Mode	US Cost	Dubai Cost	Δ Discomfort	Shift d
IPPO	Flat Shared	0.4866	0.3948	+0.050	0.105
IPPO	Local Individual	0.4161	0.3466	+0.023	0.073
IPPO	Dubai-Weighted	0.4785	0.3922	+0.036	0.093
MAPPO	Flat Shared	0.4558	0.4549	+0.023	0.023
MAPPO	Local Individual	0.4310	0.4051	+0.016	0.031
MAPPO	Dubai-Weighted	0.4304	0.4086	+0.024	0.032

The second clear finding is algorithmic. The average displacement of the IPPO conditions is 0.0904, while the average displacement of the MAPPO conditions is only 0.0286. This means that MAPPO preserves the original cost-discomfort geometry about $3.16\times$ better than IPPO under Dubai transfer. The smallest shift is observed for MAPPO with the flat shared reward, where cost changes only marginally between climates while discomfort increases by a comparatively small amount.

This is an important positive result for CTDE. MAPPO does not eliminate the comfort penalty under climate transfer, but it reduces the structural movement of the Pareto frontier. By contrast, the IPPO conditions shift much farther in cost–discomfort space, indicating greater sensitivity to the new climate. The Pareto analysis therefore provides one of the strongest quantitative arguments in favor of MAPPO in this study.

The reward-design dimension shows a weaker but still interpretable pattern. Across both algorithms, the local individual reward tends to produce the smallest increase in discomfort, while the flat shared reward often shows the largest displacement. The Dubai-weighted reward usually falls between these two. This suggests that assigning more direct local responsibility for discomfort can help preserve comfort behavior under transfer, even when it does not fully solve the underlying reward-misalignment problem.

The main implication of the Pareto analysis is that climate transfer changes the shape of the operational trade-off space. Under the US training climate, the different reward designs occupy meaningfully different trade-off positions. Under Dubai deployment, those options move toward a narrower low-cost, high-discomfort region. This supports the central claim of the paper: the problem is not simply that a policy performs worse, but that the reward function continues to optimize a trade-off that no longer corresponds to the deployment reality.

V. Results and Analysis

A. Baseline Performance (Single-Seed)

The initial single-seed experiment was used as a baseline to verify that the CityLearn MARL pipeline, reward function, observation wrapper, PPO update logic and evaluation procedure were functioning correctly under the original US climate. In this stage, IPPO and MAPPO were trained and evaluated in the same climate setting, so the experiment represents an in-distribution control problem rather than a transfer problem.

Under the fixed baseline setup, the neutral CityLearn controller achieved a mean evaluation reward of -129.14. IPPO achieved a mean reward of -130.91, which is only around 1.37% worse than the neutral baseline. This indicates that IPPO learned a policy close to the baseline controller and was able to maintain relatively stable behavior during evaluation. In contrast, MAPPO achieved a much lower mean reward of -158.43, approximately 22.68% worse than neutral, showing that the centralized critic did not improve performance in this initial setting.

The KPI results reveal a more nuanced picture. IPPO slightly worsened cost and carbon emissions compared with the neutral baseline, but it improved ramping and maintained almost identical discomfort. This suggests that IPPO learned a relatively balanced control policy, although it did not clearly outperform the passive baseline. MAPPO, however, showed stronger trade-offs: it improved discomfort and load-factor-related metrics, but at the cost of substantially worse cost, carbon emissions, and ramping. Therefore, the single-seed baseline shows that the learning framework is operational, but also highlights that reward optimization does not automatically translate into better overall KPI performance.

This baseline experiment motivated the later reward-design study. In particular, the weak MAPPO result and the close-but-not-superior IPPO result suggested that the reward formulation and training paradigm strongly affect the trade-off between cost, emissions, comfort, and grid stability. However, this performance is not well balanced. The baseline tends to prioritize cost reduction at the expense of thermal comfort and system stability, highlighting a key limitation of the default reward formulation.

Table V shows the KPI trade-offs relative to the neutral baseline across the three-seed evaluation. Although IPPO with local individual rewards achieves the closest reward performance to the neutral baseline, it still increases cost and carbon emissions slightly. However, it maintains comfort, ramping, and load-factor-related KPIs close to or better than neutral behavior.

These results show that the best learned controller does not dominate the neutral baseline across all KPIs. Instead, it produces a trade-off: slightly worse cost and carbon performance, but comparable or improved comfort, ramping, and load-shaping behavior. This motivates the multi-seed stability analysis in the next section.

B. Multi-Seed Evaluation and Stability

To reduce the effect of stochastic variation in MARL training, the main experiments were repeated across three independent seeds. Each condition was trained for 250,000 environment steps and the final results are reported as mean and standard deviation across seeds. This allows the comparison to focus on systematic differences between algorithms and reward designs rather than on a single lucky or unlucky run.

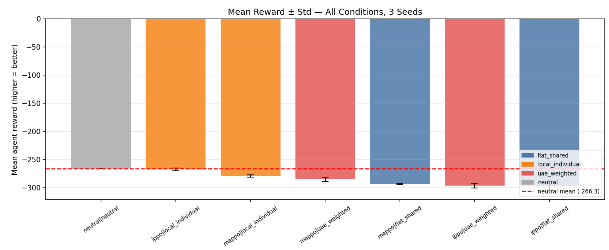

Fig. 5. Mean reward across all configurations with standard deviation over three seeds.

As shown in Fig. 5, the best-performing learned controller was IPPO with the local individual reward. It achieved a mean reward of -267.79 ± 2.33, compared with -266.34 for the neutral baseline. Although this does not outperform the neutral controller, it is the closest learned policy to the baseline and shows low variance across seeds. This indicates that decentralized training with local reward assignment is the most stable configuration among the tested MARL methods.

The remaining configurations performed worse. MAPPO with local individual reward achieved -279.57 ± 2.05, while MAPPO with Dubai-weighted reward achieved -285.30 ± 4.04. The flat shared reward variants performed poorly for both algorithms, especially IPPO, where the standard deviation was also higher. This suggests that shared rewards make credit assignment more difficult for independent agents, while local reward signals provide a clearer learning objective.

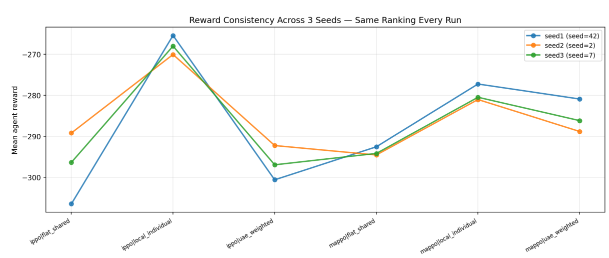

Fig. 6. Consistency of reward ranking across seeds.

Fig. 6 further supports this interpretation. The relative ranking of reward designs remains broadly consistent across the three seeds, with IPPO local individual reward repeatedly emerging as the strongest learned-controller configuration. This consistency suggests that the observed performance differences are not random artifacts of initialization. Instead, they reflect systematic differences in how each reward design shapes the learning process.

However, the multi-seed results also show an important limitation: none of the learned MARL controllers clearly

TABLE V
KPI PERCENTAGE CHANGE RELATIVE TO THE NEUTRAL BASELINE ACROSS THREE SEEDS. POSITIVE VALUES INDICATE WORSE PERFORMANCE, WHILE NEGATIVE VALUES INDICATE IMPROVEMENT.

Algorithm	Reward Design	Cost (%)	Carbon (%)	Discomfort (%)	Ramping (%)	Load Factor (%)
IPPO	Flat shared	$+14.5 \pm 5.6\%$	$+13.7 \pm 3.8\%$	$-1.5 \pm 1.4\%$	$+10.3 \pm 3.2\%$	$-2.3 \pm 0.2\%$
IPPO	Local individual	$+4.8 \pm 2.0\%$	$+5.9 \pm 1.7\%$	$-0.1 \pm 0.0\%$	$-0.2 \pm 2.0\%$	$-3.0 \pm 0.7\%$
IPPO	UAE weighted	$+12.6 \pm 5.5\%$	$+11.6 \pm 3.5\%$	$-0.8 \pm 0.7\%$	$+10.6 \pm 3.4\%$	$-2.6 \pm 0.3\%$
MAPPO	Flat shared	$+12.8 \pm 0.8\%$	$+12.5 \pm 1.2\%$	$-1.0 \pm 0.1\%$	$+10.1 \pm 0.9\%$	$-3.9 \pm 0.6\%$
MAPPO	Local individual	$+7.8 \pm 0.9\%$	$+7.9 \pm 1.0\%$	$-0.2 \pm 0.1\%$	$+2.3 \pm 1.9\%$	$-3.9 \pm 1.2\%$
MAPPO	UAE weighted	$+8.4 \pm 1.3\%$	$+8.2 \pm 1.8\%$	$-0.4 \pm 0.1\%$	$+6.9 \pm 1.1\%$	$-3.9 \pm 1.1\%$

outperform the neutral baseline in total reward. The best configuration, IPPO local individual, comes close, but still remains slightly below neutral. Therefore, the main conclusion is not that MARL universally improves CityLearn performance, but that local individual reward design produces the most reliable and balanced learned controller among the tested alternatives.

C. Overall KPI Performance (US and Dubai)

Fig. 7 compares the mean KPI values across algorithms and reward designs in both the US training climate and the Dubai transfer climate. The results show that climate transfer does not affect all KPIs in the same direction. Instead, the learned controllers exhibit a clear shift in the cost–comfort trade-off.

Across most configurations, cost and carbon emissions are lower under Dubai evaluation than under the US climate. For example, IPPO with local individual reward reduces cost from 0.4161 in the US climate to 0.3466 in Dubai, and carbon emissions from 0.4487 to 0.3717. At first glance, this appears to indicate improved transfer performance. However, this improvement is accompanied by worse thermal comfort. For the same IPPO local individual controller, discomfort increases from 0.9710 to 0.9942. A similar pattern appears across the other reward designs.

This result suggests that climate transfer changes the operational trade-off rather than uniformly degrading performance. Under Dubai conditions, the agents are able to reduce grid-related KPIs such as cost and carbon, but they do so while allowing comfort performance to deteriorate. This is important because the reward signal alone may make the controller appear effective, while the KPI breakdown reveals that the policy is not balancing objectives in the same way as it did in the training climate.

TABLE VI
OVERALL KPI COMPARISON ACROSS REWARD DESIGNS AND ALGORITHMS UNDER US AND DUBAI CLIMATES.

Method	Climate	Cost	Discomfort	Composite Δ
IPPO (Flat)	US	0.4866	0.9433	-
IPPO (Flat)	Dubai	0.3948	0.9933	-7.94%
IPPO (Local)	US	0.4161	0.9710	-
IPPO (Local)	Dubai	0.3466	0.9942	-6.67%
IPPO (Dubai-weighted)	US	0.4785	0.9571	-
IPPO (Dubai-weighted)	Dubai	0.3922	0.9929	-7.09%
MAPPO (Flat)	US	0.4558	0.9636	-
MAPPO (Flat)	Dubai	0.4549	0.9863	-3.99%
MAPPO (Local)	US	0.4310	0.9719	-
MAPPO (Local)	Dubai	0.4051	0.9884	-3.87%

Table VI summarizes this trade-off using cost, discomfort and a composite change measure. The composite values are negative because cost decreases more strongly than discomfort increases. However, this should not be interpreted as a complete improvement under Dubai conditions. Since thermal discomfort consistently increases, the transfer result is better understood as a shift toward lower-cost but less comfortable operation. This indicates that agents exploit the environment to reduce energy consumption, but fail to properly adapt to increased cooling demand, resulting in poorer comfort.

Figure 8 further supports this interpretation. The KPI heatmap shows that methods with lower cost are not necessarily the most balanced across all objectives. IPPO with local individual reward remains the strongest overall configuration because it maintains the lowest cost among the learned controllers while keeping discomfort close to the neutral baseline. In contrast, some MAPPO configurations improve load-shaping metrics but do not achieve the same cost–comfort balance.

Overall, the KPI results show that reward design affects not only the magnitude of performance, but also the type of behavior learned by each controller. A controller that performs well on cost and carbon may still be undesirable if it sacrifices occupant comfort. Therefore, the Dubai transfer evaluation should be interpreted through the full KPI profile rather than through reward or cost alone.

Thus, the Dubai evaluation reveals a reward-transfer mismatch: policies can improve cost-related KPIs while simultaneously worsening comfort, meaning that aggregate reward or composite metrics alone are insufficient for judging transfer success.

D. Comparison of Reward Design

Different reward formulations lead to distinct performance profiles. As observed in Fig. 7, flat shared rewards encourage cooperative behavior but do not explicitly penalize discomfort, leading to weaker comfort performance.

The three reward designs produce clearly different control behaviours. The flat shared reward performs poorly for IPPO because all agents receive the same district-level reward, making credit assignment difficult for independent learners. This is reflected in the multi-seed results, where IPPO with flat shared reward has higher cost and carbon emissions than the local individual variant.

The local individual reward provides the strongest overall result. Across three seeds, IPPO with local individual reward achieves the closest reward performance to the neutral baseline and the most balanced KPI profile. It increases cost by only +4.8 ± 2.0% and carbon emissions by +5.9 ± 1.7%,

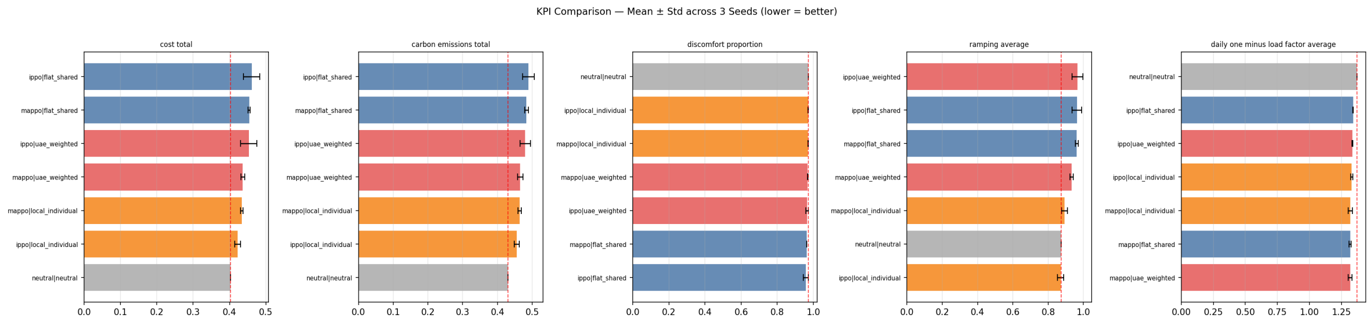

Fig. 7. KPI comparison across all configurations (mean ± std over three seeds).

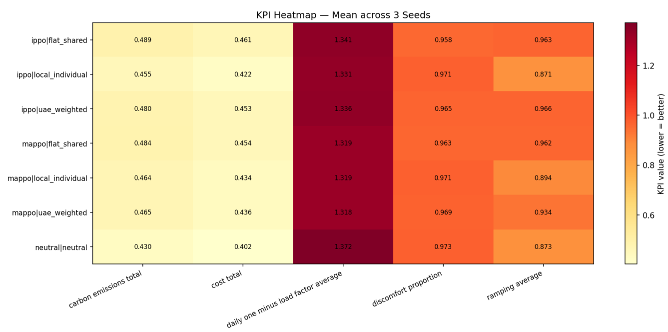

Fig. 8. Heatmap summarizing average KPI performance across all methods.

while maintaining discomfort, ramping and load-factor-related metrics close to or better than neutral behavior.

The Dubai-weighted reward does not consistently improve performance. Although it was designed to place greater emphasis on climate-relevant objectives, it generally produces higher cost and carbon emissions than the local individual reward. This suggests that simply reweighting the reward is not sufficient, the structure of reward assignment also matters. In this experiment, local per-building credit assignment is more important than heavier KPI weighting.

E. Effect of Reward Design on System Behavior

Reward design directly influences agent decision-making. The patterns in Fig. 8 show that cost-focused rewards lead to aggressive energy-saving strategies, often at the expense of comfort.

In contrast, balanced rewards produce smoother control behavior and improved system stability, highlighting the importance of proper reward shaping.

Reward design changes the type of behavior learned by the agents. Flat shared rewards encourage district-level cooperation, but they weaken the ability of each agent to identify which of its own actions improved or worsened performance. As a result, the learned policies are less balanced and show larger penalties in cost, carbon emissions, and ramping.

Local individual rewards produce more stable behavior because each building receives feedback directly linked to its own energy use, comfort and ramping contribution. This makes the learning signal clearer, especially for IPPO, where each building is trained independently.

The UAE-weighted reward shifts the policy toward comfort and load-shaping objectives, but this comes at the expense of cost and carbon performance. Therefore, the results show that reward shaping creates trade-offs rather than universal improvement. A reward that improves one KPI can worsen another, so performance must be evaluated using the full KPI profile rather than reward alone.

F. IPPO vs MAPPO Comparison

As seen in Fig. 7, IPPO demonstrates more stable and consistent performance across configurations. MAPPO, while competitive in some KPIs, exhibits higher variability, suggesting that centralized training does not necessarily improve robustness

Across the three 250k-step seeds, IPPO is more reliable than MAPPO. The best learned controller is IPPO with local individual reward, achieving a mean reward of -267.79 ± 2.33, compared with -266.34 for the neutral baseline. The best MAPPO configuration, MAPPO with local individual reward, achieves -279.57 ± 2.05, which is noticeably worse.

This suggests that centralized training does not inherently improve robustness under distributional shift. Although MAPPO has access to the global state through its centralized critic during training, this does not translate into better evaluation performance. One likely explanation is that the centralized critic must learn a more complex value function over the full district state, while IPPO benefits from simpler local credit assignment.

Therefore, the results support the conclusion that decentralized IPPO with local reward assignment is the strongest and most robust learning approach among the tested configurations.

G. Cross-Climate Generalization (US → Dubai)

Figure 9 shows that when evaluated under Dubai conditions, policies reduce cost but significantly increase discomfort.

The Dubai transfer evaluation tests whether policies trained under the US climate can generalize without retraining. The results show that transfer does not cause uniform degradation across all KPIs. Instead, it changes the trade-off between energy efficiency and comfort.

Under Dubai conditions, cost and carbon emissions often decrease relative to the US evaluation. For example, IPPO

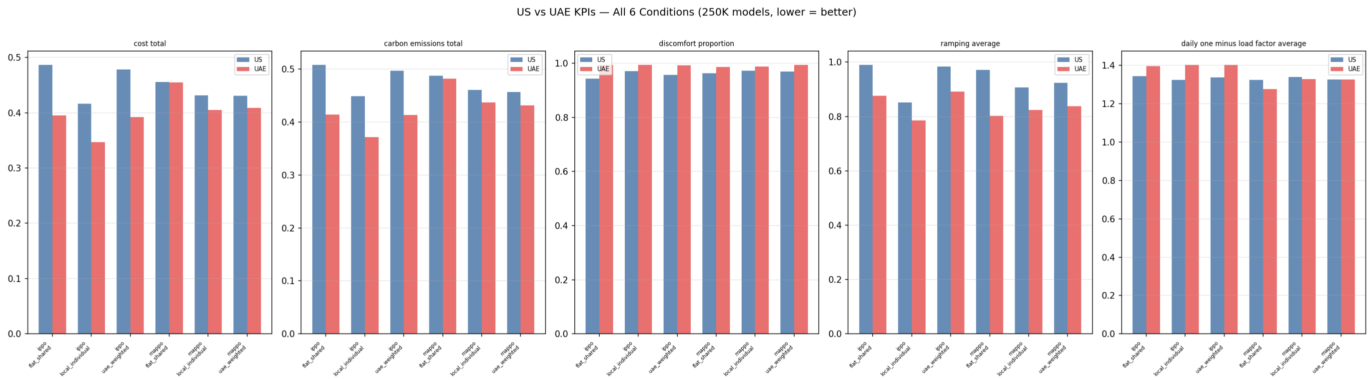

Fig. 9. KPI comparison under US (training) and Dubai (evaluation) climates.

with local individual reward reduces cost from 0.4161 to 0.3466 and carbon emissions from 0.4487 to 0.3717. However, discomfort increases from 0.9710 to 0.9942. This indicates that the controller can reduce energy-related KPIs under the Dubai dataset, but it does so while allowing comfort to deteriorate.

This result is important because it shows that climate transfer cannot be judged using cost or reward alone. A controller may appear to perform better on energy metrics while simultaneously becoming less acceptable from an occupant-comfort perspective.

H. KPI Degradation Under Climate Transfer

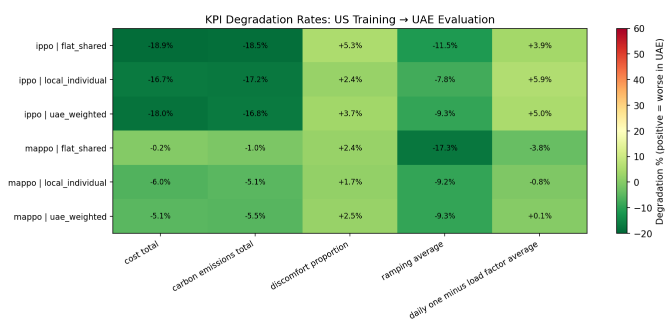

Fig. 10. KPI degradation when transferring from US to Dubai climate.

As illustrated in Fig. 10, discomfort increases by up to +5.3%, while cost decreases by up to -18.9%.

This means that the learned policies exploit the new environment in a way that reduces energy consumption, but they do not maintain the same comfort balance. The transfer problem is therefore not simply that all KPIs become worse. Instead, the policy shifts toward lower-cost operation at the expense of thermal comfort.

This supports the central argument of the project: reward functions trained under one climate may not preserve the same KPI priorities when deployed in another climate.

I. Effect of Reward Design on Transfer Degradation

Based on Fig. 10, local and weighted rewards show slightly improved robustness compared to flat rewards. However, none of the configurations fully eliminate the performance gap.

Reward design affects how strongly each controller shifts under climate transfer. Local and UAE-weighted rewards show slightly better robustness than flat shared rewards, but none of the reward designs fully eliminates the cost-comfort trade-off.

The flat shared reward generally produces larger transfer shifts because the shared reward does not give agents precise feedback about their individual contribution. Local individual rewards reduce this problem by giving each building a more direct learning signal. UAE-weighted rewards improve some climate-relevant metrics, but they do not consistently outperform local individual rewards.

Overall, the transfer results show that reward shaping alone is not enough to guarantee robustness. Future work should combine improved reward design with climate-aware training, domain randomization, or retraining on target-climate data.

J. IPPO vs MAPPO Under Climate Transfer

Under Dubai conditions, as observed in Fig. 9, IPPO maintains more stable performance, while MAPPO becomes more sensitive to environmental changes.

The climate-transfer results further confirm the difference between IPPO and MAPPO. Under the US training climate, IPPO with local individual reward is the strongest learned controller, achieving reward performance closest to the neutral baseline. Under Dubai evaluation, IPPO also maintains a more balanced KPI profile than MAPPO.

MAPPO does not show a clear transfer advantage despite using a centralized critic during training. In several configurations, MAPPO improves some load-shaping or discomfort-related metrics, but this comes with higher cost and carbon emissions. For example, under the multi-seed US evaluation, MAPPO local individual reward increases cost by +7.8% and carbon emissions by +7.9% relative to neutral, while IPPO local individual increases cost by only +4.8% and carbon emissions by +5.9%.

This suggests that centralized training does not necessarily improve robustness under climate shift. The centralized critic may help coordination in theory, but in this setting it appears more sensitive to reward design and environmental distribution shift. IPPO benefits from simpler local credit assignment, making it more stable across both training and transfer evaluations. text

K. Reward Transfer Score Results

The combined observations from Fig. 7 and Fig. 10 confirm that no reward configuration dominates across all KPIs, reinforcing the multi-objective nature of the problem.

The Reward Transfer Score was introduced to test whether changes in reward reliably reflect changes in operational KPIs under climate transfer. If reward transfer were well aligned, reward changes from US to Dubai would correlate strongly with KPI changes.

However, the computed Reward Transfer Score was close to zero:

$$RTS = -0.0226, p = 0.9662.$$

This indicates that there is no meaningful statistical relationship between reward change and KPI change across the tested configurations. In other words, a policy can appear better or worse according to reward while showing a different pattern across the actual CityLearn KPIs.

This result supports one of the central claims of the study: reward functions calibrated in one climate do not necessarily remain reliable indicators of performance in another climate. Under Dubai transfer, reward alone is not sufficient for evaluating controller quality. The full KPI profile must be considered.

L. Seasonal Regime Breakdown

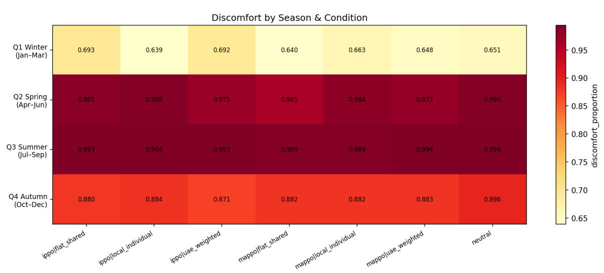

Fig. 11. Seasonal variation in discomfort.

Figure 11 shows that comfort performance varies strongly across climate regimes. As shown in Fig. 11, discomfort is lowest during the cooler winter period and increases during hotter periods, especially in the summer regime. This confirms that the transfer problem is not uniform across time; the learned controllers are most challenged when cooling demand is sustained.

This pattern is important because a policy may perform acceptably on average while still producing poor comfort outcomes during the most demanding periods. In energy management, these high-stress periods are operationally important because they coincide with peak cooling loads and increased risk of comfort violations.

The seasonal results therefore show that climate robustness cannot be evaluated only using annual average KPIs. A controller that appears stable overall may still fail during extreme seasonal regimes. This supports the need for climate-aware training or seasonal reward adaptation. text

M. Pareto Frontier Analysis

The Pareto frontier analysis examines the trade-off between cost and discomfort. As shown in Fig. 12, the relationship between these objectives changes between the US and Dubai climates. In general, points with lower cost tend to correspond to higher discomfort, showing that the controllers reduce energy use partly by relaxing comfort performance.

This trade-off is especially important under Dubai conditions. The hotter climate increases cooling demand, making it harder to reduce cost while maintaining comfort. As a result, the Pareto frontier shifts, indicating that the same reward design does not produce the same cost-comfort balance after climate transfer.

The Pareto analysis reinforces the broader conclusion that no reward configuration dominates across all objectives. Instead, each design selects a different compromise between cost, comfort, and system stability. For deployment, this means that reward functions should be calibrated according to the target climate and operational priorities, rather than assumed to transfer unchanged.

These results collectively demonstrate that climate transfer affects not only performance magnitude but also the structure of decision-making trade-offs, reinforcing the need for multi-KPI evaluation.

VI. Discussion

The results provide three key insights into MARL behavior under climate transfer. First, reward design strongly affects learned control behavior. The best-performing learned controller was IPPO with local individual reward, which achieved the closest reward performance to the neutral baseline across three seeds. This suggests that direct per-building credit assignment is more effective in this setting than shared district-level reward signals.

Second, centralized training did not provide a clear advantage. Although MAPPO uses a centralized critic during training, it did not outperform IPPO in either reward performance or overall KPI balance. This indicates that access to global state information is not sufficient to guarantee improved coordination, especially when the reward signal is multi-objective and the environment changes between training and deployment.

Third, climate transfer changes the KPI trade-off rather than uniformly degrading all metrics. Under Dubai conditions, cost and carbon emissions often decrease, with cost reductions up to 18.9%. However, this comes with increased discomfort, reaching up to 5.3%. This shows that the learned policies shift toward lower-energy operation but do not preserve the same comfort balance under higher cooling demand.

The Reward Transfer Score provides further evidence of this mismatch. The RTS value was close to zero and statistically non-significant, indicating that changes in reward were not reliably correlated with changes in operational KPIs. This means that reward improvement under transfer cannot be treated as evidence of improved real-world performance. Instead, evaluation must consider the full KPI profile, particularly comfort and grid-stability metrics.

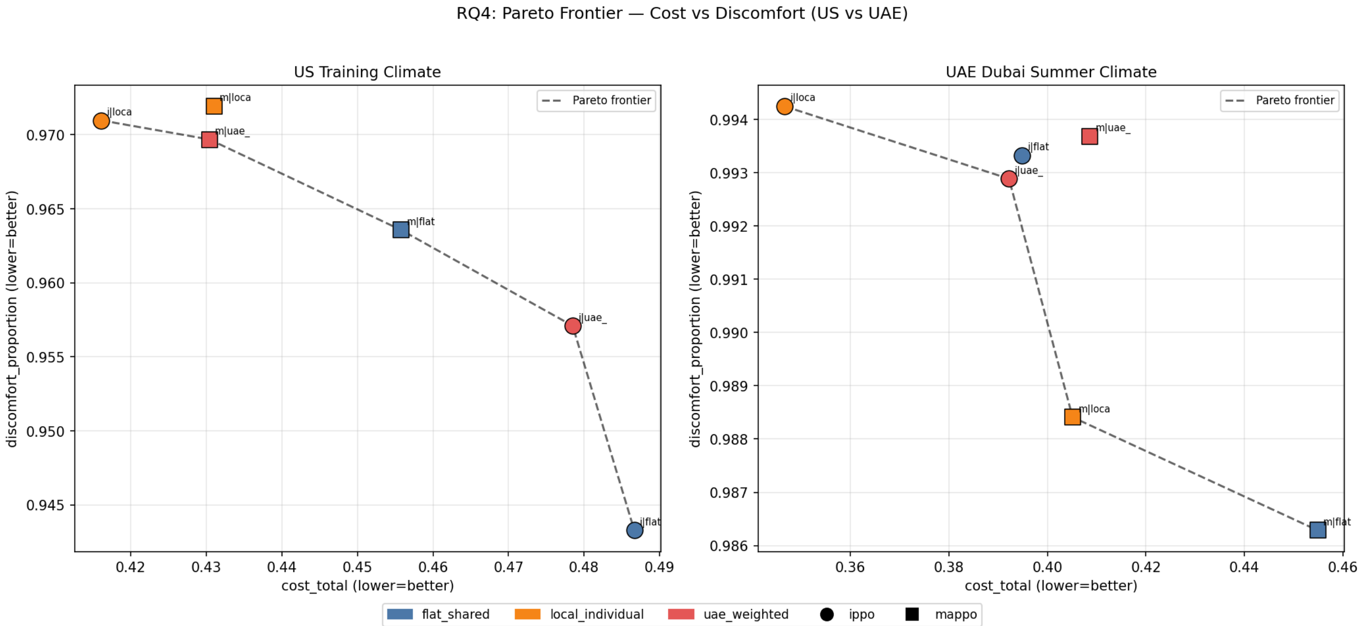

Fig. 12. Pareto frontier showing trade-offs between cost and comfort.

From a practical perspective, these findings suggest that policies trained in one climate should not be deployed in another climate without adaptation. Even when energy-related KPIs improve, comfort performance may deteriorate. This is especially important for cooling-dominated regions such as Dubai, where thermal comfort is a central operational constraint.

Overall, the study shows that reward design improves in-distribution behavior, but reward design alone is not enough to guarantee robust climate transfer. Reliable deployment requires reward functions that are calibrated to the target climate, together with environment-aware training strategies.

VII. Conclusion and Future Work

This paper investigated the role of reward design in multi-agent reinforcement learning (MARL) for urban energy control under zero-shot climate transfer. Using the CityLearn 2023 dataset and real Dubai weather data, we evaluated how policies trained in a temperate US climate behave when deployed in a fundamentally different, cooling-dominated environment. IPPO and MAPPO were analyzed across flat shared, local individual and UAE-weighted reward designs using multiple training seeds.

The results demonstrate that reward design strongly influences system behavior but does not guarantee robustness under climate transfer. Across all configurations, a consistent pattern emerges: cost and carbon emissions decrease, while thermal discomfort increases. In some cases, this corresponds to significant cost reductions (up to 18.9%) accompanied by increases in discomfort (up to 5.3%). This indicates that learned policies adapt to reduce energy consumption but fail to adapt to increased cooling demand.

More importantly, the findings reveal that the challenge is not merely performance degradation, but a structural shift in optimization behavior. Pareto frontier analysis shows that all methods move toward a low-cost, high-discomfort region under Dubai conditions, indicating that the cost–comfort trade-off itself changes under climate transfer. Seasonal regime analysis further highlights that this effect is not uniform: policies degrade significantly under extreme summer conditions due to demand saturation, while partially recovering during milder periods where the target climate overlaps with the training distribution.

The Reward Transfer Score (RTS) provides strong evidence of reward misalignment. Near-zero RTS values indicate that reward signals become effectively decoupled from actual KPI outcomes under climate shift. This confirms the central claim of this work: reward misalignment, rather than model limitations, is a primary driver of performance degradation when deploying MARL controllers across climates.

From an algorithmic perspective, IPPO with local individual reward emerges as the most robust configuration, achieving the closest performance to the neutral baseline and the most balanced KPI profile. MAPPO does not provide a consistent advantage, suggesting that centralized training alone is insufficient to improve robustness under distributional shift.

In summary, the study answers the research questions as follows. First, climate transfer primarily increases thermal discomfort despite reductions in cost and emissions. Second, reward shaping influences performance but does not prevent degradation, instead shifting the trade-off structure. Third, decentralized IPPO demonstrates greater robustness than MAPPO. Fourth, climate transfer significantly alters the cost–comfort Pareto frontier.

Overall, these results show that policies trained in one climate cannot be directly deployed in another without adaptation. Even when reward optimization appears successful, it may correspond to undesirable system behavior, particularly in extreme environments such as Dubai. This highlights the importance of multi-KPI evaluation and environment-aware validation in real-world MARL deployment.

Future work should focus on improving generalization and robustness of MARL controllers under environmental shift. First, climate-aware training strategies such as domain randomization or multi-climate training should be explored to expose agents to diverse weather regimes during learning. Second, adaptive reward design mechanisms, including dynamic or context-aware weighting of objectives, can help align reward signals with environment-specific operational priorities.

Third, transfer learning approaches, including fine-tuning on target-climate data and meta-learning for rapid adaptation, offer a pathway to bridging the distribution gap between training and deployment environments. Fourth, incorporating uncertainty modeling and risk-sensitive optimization may improve robustness under extreme and unpredictable conditions. Additionally, integrating constraint-aware or safe MARL formulations can ensure that critical objectives such as thermal comfort are not violated during deployment.

Finally, extending this analysis to larger building districts and additional climate regions will help determine whether the observed reward misalignment and trade-off shifts are general phenomena in urban energy MARL systems. These directions aim to move MARL-based energy control beyond benchmark performance toward reliable, deployment-ready solutions in real-world smart city environments.

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